THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 12	Let the point be $(0, y)$
2	**(d)** 63	Twice the area
3	**(b)** $\frac{16}{3}$	Equate squared distances
4	**(a)** 6	$x = 0$: $mc + na = 0$
5	**(c)** 50	$s^2 = d^2/2$
6	**(d)** 28	Use the area formula
7	**(a)** 9	Let the point be $(0, y)$
8	**(b)** 44	Twice the area
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\sqrt{9+16} = 5$; R is the formula used.
10	**(c)** A is true but R is false.	$\left(\frac{2+8}{2}, \frac{-3+5}{2}\right) = (5, 1)$: A true; the formula uses sums, not differences, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
12	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 1(8-k) + 2(k-5) + 4(5-8)\rvert = 0$ (1); $k = 14$ (1).	1+1+1
13	Midpoint of the opposite side $= (3,\ 4)$ (1); length $= 5$ (1).	1+1
14	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
16	P divides AB in $1:2$: $(3,\ 2)$ (1.5); Q divides AB in $2:1$: $(6,\ 4)$ (1.5).	1.5+1.5
17	$(6)^2 + (y-3)^2 = 100$ (1); $(y-3)^2 = 64$ (1); $y = 3 \pm 8$, i.e. $y = 11$ or -5 (1).	1+1+1
	OR: $AB^2 = 100$, $BC^2 = 100$, $CD^2 = 100$, $DA^2 = 100$ (2); all four sides equal (1); diagonals $AC^2 = 320$, $BD^2 = 80$ differ so it is a rhombus, not a square (2).	2+1+2
18	P divides AB in $1:2$: $(6,\ 5)$ (1.5); Q divides AB in $2:1$: $(9,\ 7)$ (1.5).	1.5+1.5
	OR: $(12)^2 + (y-1)^2 = 169$ (1); $(y-1)^2 = 25$ (1); $y = 1 \pm 5$, i.e. $y = 6$ or -4 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$ (1). (ii) $SL = \sqrt{6^2 + 8^2} = 10$ units (1). (iii) $LH = \sqrt{(8-2)^2 + 0} = 6$ units (1). (iv) $SH = 8$; $SH^2 + LH^2 = 64 + 36 = 100 = SL^2$, so by the converse of Pythagoras the triangle is right-angled at H (2).	1+1+1+2
20	Centroid $= (4,\ 4)$ (2); area $= \frac{1}{2}\lvert \ldots \rvert = 18$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $\left(\frac{11}{2}, 7\right)$ (2); this is also the midpoint of BD (1); $D = (4,\ 11)$ (2).	2+1+2

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 50	$s^2 = d^2/2$
2	**(c)** 28	Use the area formula
3	**(c)** $\frac{22}{3}$	Let the point be $(0, y)$
4	**(b)** 78	Twice the area
5	**(b)** $\frac{17}{5}$	Equate squared distances
6	**(b)** 10	$x = 0$: $mc + na = 0$
7	**(d)** 50	$s^2 = d^2/2$
8	**(d)** 30	Use the area formula
9	**(d)** A is false but R is true.	$(0, 5)$ has x-coordinate 0 so it is on the y-axis: A false; R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$\left(\frac{1\cdot4+2\cdot1}{3}, \frac{1\cdot5+2\cdot2}{3}\right) = (2, 3)$ so A is true; R is true but unrelated to the section formula.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area of triangle must be 0 (1); $\frac{1}{2}\lvert -2(7-k) + 0(k-3) + 3(3-7)\rvert = 0$ (1); $k = 13$ (1).	1+1+1
12	Midpoint of the opposite side $= (5,\ 6)$ (1); length $= 5$ (1).	1+1
13	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
14	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 2(1-k) + 4(k+3) + 5(-3-1)\rvert = 0$ (1); $k = 3$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
16	P divides AB in $1:2$: $(0,\ 4)$ (1.5); Q divides AB in $2:1$: $(3,\ 2)$ (1.5).	1.5+1.5
17	$(8)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 36$ (1); $y = 2 \pm 6$, i.e. $y = 8$ or -4 (1).	1+1+1
	OR: $AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 90$, $BD^2 = 10$ differ so it is a rhombus, not a square (2).	2+1+2
18	P divides AB in $1:2$: $(4,\ 1)$ (1.5); Q divides AB in $2:1$: $(7,\ 4)$ (1.5).	1.5+1.5
	OR: $(12)^2 + (y+0)^2 = 169$ (1); $(y+0)^2 = 25$ (1); $y = 0 \pm 5$, i.e. $y = 5$ or -5 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $P = \left(\frac{1(10)+2(-2)}{3}, \frac{1(13)+2(1)}{3}\right) = (2, 5)$ (1). (ii) $(4, 7)$ (1). (iii) $Q = \left(\frac{2(10)+1(-2)}{3}, \frac{2(13)+1(1)}{3}\right) = (6, 9)$ (1). (iv) $AP = \sqrt{16+16} = 4\sqrt{2}$, $PQ = \sqrt{16+16} = 4\sqrt{2}$, $QB = \sqrt{16+16} = 4\sqrt{2}$; all equal (2).	1+1+1+2
20	Centroid $= (6,\ 6)$ (2); area $= \frac{1}{2}\lvert \ldots \rvert = 27$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $(7, 2)$ (2); this is also the midpoint of BD (1); $D = (6,\ 4)$ (2).	2+1+2

Mistakes to avoid:

- Using $(x_1 - x_2)$ and $(y_2 - y_1)$ inconsistently or forgetting to square a negative difference.
- Leaving the square root out, giving AB^2 instead of AB.
- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
- Using the mid-point formula with differences instead of sums.
- Writing a point on the x-axis as $(0, y)$; it is $(x, 0)$.
- Declaring a triangle right-angled without checking that the *longest* side squared equals the sum of the other two.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Coordinate Geometry** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	Equate squared distances
2	**(d)** 12	$x = 0$: $mc + na = 0$
3	**(d)** 100	$s^2 = d^2/2$
4	**(a)** 42	Use the area formula
5	**(b)** $-\frac{9}{2}$	Let the point be $(0, y)$
6	**(d)** 60	Twice the area
7	**(a)** 10	Equate squared distances
8	**(b)** 9	$x = 0$: $mc + na = 0$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Sides are 4, 3 and $\sqrt{16+9} = 5$, and $5^2 = 4^2 + 3^2$; R is the converse of the Pythagoras theorem used.
10	**(c)** A is true but R is false.	x is negative and y is positive in the second quadrant, so A is true and R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Midpoint of the opposite side $= (4,\ 5)$ (1); length $= 5$ (1).	1+1
12	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
13	Area of triangle must be 0 (1); $\frac{1}{2}\lvert 3(8-k) + 6(k-2) + 9(2-8)\rvert = 0$ (1); $k = 14$ (1).	1+1+1
14	Midpoint of the opposite side $= (6,\ 4)$ (1); length $= 5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
16	P divides AB in $1:2$: $(5,\ 3)$ (1.5); Q divides AB in $2:1$: $(8,\ 1)$ (1.5).	1.5+1.5
17	$(8)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 36$ (1); $y = 2 \pm 6$, i.e. $y = 8$ or -4 (1).	1+1+1
	OR: $AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
18	P divides AB in $1:2$: $(4,\ 5)$ (1.5); Q divides AB in $2:1$: $(8,\ 7)$ (1.5).	1.5+1.5
	OR: $(8)^2 + (y+1)^2 = 289$ (1); $(y+1)^2 = 225$ (1); $y = -1 \pm 15$, i.e. $y = 14$ or -16 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(x, 0)$ (1). (ii) $PA^2 = (x-2)^2 + 25$, $PB^2 = (x+2)^2 + 81$ (1). (iii) $PA^2 = PB^2 \Rightarrow -4x + 29 = 4x + 85 \Rightarrow x = -7$; $P(-7, 0)$ (1). (iv) $PA = \sqrt{81+25} = \sqrt{106}$ units; $PB = \sqrt{25+81} = \sqrt{106}$ units (2).	1+1+1+2
20	Centroid $= (1,\ 2)$ (2); area $= \frac{1}{2}\lvert \ldots \rvert = 18$ sq. units (3).	2+3
	OR: Diagonals bisect each other: midpoint of AC is $(7, 8)$ (2); this is also the midpoint of BD (1); $D = (5,\ 12)$ (2).	2+1+2

Mistakes to avoid:

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- In the section formula, swapping the ratio: the multiplier of x_2 is m_1 (not m_2).
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